MIL-S-50482A (MU)
20 December 1971
SUPERSEDING
MIL-S-50482 (MU)
15 January 1970

MILITARY SPECIFICATION

SIGNAL, ILLUMINATION, GROUND: YELLOW, XM191, GREEN, XM192 and RED, XM193 PARTS FOR AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for and the loading, assembling and packing for one type of signal designated as Signal, Illumination, Ground, XM191 (Yellow), XM192 (Green), and XM193 (Red).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	-	Preservation, Method of
MIL-A-2550	-	Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

FSC: 1370

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MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-286	-	Propellant, Solid, Sampling Examination and Testing
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1234	-	Pyrotechnics, Sampling, Inspection and Testing
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

9233945	-	Signal, Illumination, Ground Assembly
9242362	-	Bag, Packing, Ammunition, for Signal, Illumination, Ground, XM191, XM192, XM193
9242363	-	Box, Packing, Ammunition, For Signal, Illumination Ground, XM191, XM192, XM193

PUBLICATIONS

ET-9233945	-	Equipment Tabulation Number

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly.-The assembly shall comply with all requirements specified on Drawing (Dwg.) 9233945 and with all requirements specified in applicable specifications.

3.3 Functioning.-The signal shall ignite and comply with the following requirements when tested as specified in 4.4.3.

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3.3.1 Safety.-The signal shall not explode at anytime during functioning.

3.3.2 Visibility.-The signal color shall be visible on a clear dark night from the air at a minimum slant range of one mile and a minimum altitude of one thousand (1000) feet.

3.4 Leakage of bag.-The packed, sealed barrier bag shall exhibit no evidence of leakage, when subjected to the Quick Leak Test as specified in 4.4.4.

3.5 Workmanship.-All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, crazes, porosity, warpage, burn marks, blisters, excess flash, scratches, dirt, grease, rust, corrosion products and other defects and foreign matter which would affect appearance or serviceability. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined. All packing components shall be dry.

3.6 First article inspection.-This specification makes provisions for the first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of finding on the moisture content of the first fire composition, illuminant composition and paper and paper products shall be furnished with each lot submitted for acceptance.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.-Prior to the start of regular production, the contractor shall submit a first article sample as designated by the contracting officer (see 6.1c) for evaluation in accordance with the provisions of 4.2.1 and 4.2.2. The first article sample shall consist of inert parts, subassemblies and loaded assemblies in the quantities specified herein, which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing components shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the

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contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including statements of findings and certificates of conformance of materials. A sample containing known defects shall not be submitted unless specifically authorized by the contracting officer. A first article sample or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit an additional sample or portion thereof of items for first article inspection. Unless otherwise directed by the contracting officer, loaded components shall not be submitted for proving ground testing until all other phases of the first article samples have been accepted. Until the first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.1 Inert parts

4.2.1.1 Submission.-Twenty-five complete sets of inert parts (i.e., 25 each of every inert component and inert subassembly) shall be submitted for inspection to a Government approved laboratory designated by the contracting officer.

4.2.1.2 Inspections to be performed.-The inert parts will be subjected to any or all of the applicable examinations and tests specified in 4.3.2 and 4.3.3 of this specification and any or all of the requirements of the applicable drawings.

4.2.1.3 Rejection.-If any inert component or subassembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an inert component or subassembly to comply with any of the stated requirements.

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4.2.2 Loaded assemblies

4.2.2.1 Submission.-Six (6) packed and sealed barrier bags and twenty (20) illuminant assemblies of each designation shall be selected for tests to be conducted by the Government at the contractor's plant. Fifty (50) loaded signal assemblies of each designation shall be submitted to a Government proving ground designated by the contracting officer.

4.2.2.2 Inspections to be performed.-The following tests shall be conducted in accordance with the applicable procedures of 4.4 herein:

a. Leakage (see 4.4.4) - Six (6) packed and sealed barrier bags.
b. Candlepower (see 4.4.2) - Twenty (20) illuminant assemblies of each designation.
c. Functioning (see 4.4.3) - Fifty (50) signal assemblies of each designation (see visibility requirement, 3.3.2).

4.2.2.3 Rejections.-The first article sample shall be rejected if any of the following conditions occur:

a. If a failure occurs during the leakage test (see 3.4).
b. If any illuminant assembly fails to meet the candlepower requirements (see dwg. 9233957).
c. If any signal fails to meet the visibility requirements (see 3.3.2).
d. If a critical defect occurs or if two or more signals exhibit any of the major defects as classified in Table II.

The Government reserves the right to terminate its inspection upon encountering any failure of an item in the sample to comply with any or all requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not

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be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same material and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded signals shall contain:

a. Ignitacord from not more than one lot.
b. Composition ingredients from not more than one lot.
c. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Cup, igniter (see dwg. 9233954 covering a detail of dwg. 9233945)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 0.40 Percent		
101.	Diameter of hole, maximum (max)	Gage	01001
102.	Diameter below rim	Gage	01002
Minor:	AQL 1.00 Percent		
201.	Total length	Gage	01003
202.	Evidence of poor workmanship (See 3.6)	Visual	01004

4.3.2.2 Cup, igniter assembly (see dwg. 9233953 covering a detail of dwg. 9233945)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Composition on lip of cup	Visual	02001
Major:	AQL 0.40 Percent		
101.	Igniter charge missing	Visual	02002
Minor:	AQL 1.00 Percent		
201.	Igniter charge obstructs hole	Visual	02003
202.	Evidence of poor workmanship (See 3.6)	Visual	02004

4.3.2.3 Ignitacord Assembly (see dwg. 9233970 covering a detail of dwg. 9233945)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 Percent		
101.	Length of ignitacord	Gage	03001
102.	Length of tubing	Gage	03002
103.	Location of tubing	Gage	03003
104.	Tubing loose	Manual	03004

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Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	03005

4.3.2.4 Wire, pull assembly (see dwg. 9233950 covering a detail of dwg. 9233945)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Friction composition on last three spaces of pull wire	Visual	04001
Major:	AQL 0.40 Percent		
101.	Friction composition in designated area missing or inadequate	Visual	04002

Minor: None defined

4.3.2.5 Housing and igniter assembly (see dwg. 9233946 covering a detail of dwg. 9233945)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined			
Major:	AQL 0.40 Percent		
101.	RTV missing or inadequate	Visual	05001
102.	Component missing or incorrectly assembled	Visual/Manual	05002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	05003

4.3.2.6 Illuminant case loading assembly (see dwg. 9233957 covering a detail of dwg. 9233945)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			

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Major:	AQL 0.40 Percent		
101.	First fire composition missing	Visual	06001
102.	Illuminant composition missing	Visual	06002
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	06003

4.3.2.7 Assembly prior to sealing cover (see dwg. 9233945)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 Percent		
101.	Pellet missing	Visual	07001
102.	Spacer missing	Visual	07002
103.	Unshielded portion of ignitacord on first fire composition not covered by ignitacord disc	Visual	07003
104.	Any component missing, incorrectly located, insecure or damaged to extent that function of item may be impaired	Visual/Manual	07004
105.	Tape missing or incorrectly positioned	Visual	07006
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	07005

4.3.2.8 Assembly after sealing (see dwg. 9233945)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color marking incorrect	Visual	08001
2.	Safety clip missing or insecure	Visual/Manual	08002

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Major:	AQL 0.40 Percent		
101.	Container damaged to extent that function of item may be impaired	Visual	08003
Minor:	AQL 1.00 Percent		
201.	Marking other than color marking misleading, or unidentifiable	Visual	08004
202.	Nameplate missing	Visual	08005

4.3.2.9 Bag, packing (see dwg. 9242362)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color marking incorrect	Visual	09001
Major:	AQL 0.40 Percent		
101.	Bag ruptured, punctured or seal incomplete	Visual/Manual	09002
Minor:	AQL 2.50 Percent		
201.	Heat seal incorrectly located	Visual	09003
202.	Number of signals in bag incorrect	Visual	09004
203.	Air not removed from bag	Manual	09005
204.	Marking other than color missing, misleading or unidentifiable	Visual	09006
205.	Evidence of poor workmanship (See 3.6)	Visual	09007

4.3.2.10 Wood packing box, prior to sealing (see dwg. 9242363)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			

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Minor:	AQL 1.50 Percent		
201.	Number of bags incorrect	Visual	10001
202.	Packing component missing	Visual	10002
203.	Evidence of poor workmanship (see 3.6)	Visual	10003

4.3.2.11 Sealed wood packing box (see dwg. 9242363)

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Color marking incorrect	Visual	11001
Major:	AQL 0.40 Percent		
101.	Box damaged to extent that contents are exposed or liable to become exposed	Visual	11002
102.	Strapping missing, broken or loose	Visual/Manual	11003
Minor:	AQL 1.50 Percent		
201.	Hardware or strapping improperly assembled or engaged	Visual	11004
202.	Car seal unsealed or incorrectly positioned	Visual	11005
203.	Contents loose	Manual	11006
204.	Marking other than color misleading or unidentifiable	Visual	11007

4.3.3 Testing

4.3.3.1 Moisture Content (see Table I), Major Defect

TABLE I

Material	Code No.
First fire composition (see dwg. 9233957)	06004
Illuminant composition (see dwg. 9233957)	06005
Felt Paper & Paper Products (see dwg. 9242362 & dwg. 9233986)	06006

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The contractor shall provide adequate controls to insure that the materials comply with the requirement. For verification, the contractor shall select and subject to test, one sample of each material from each eight hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excess moisture, the remaining unloaded material and the loaded components shall be rejected. Test procedures shall be as specified in 4.4.1.

4.3.3.2 Candlepower (see dwg. 9233957), Major defect, Code No. 06007.-Twenty (20) illuminant assemblies shall be selected from each lot for test. If two (2) or more assemblies fail to comply with the requirement, the lot shall be rejected. The test shall be performed as specified in 4.4.2.

4.3.3.3 Functioning (see 3.3 and Table II).-The signal assembly shall be observed for any evidence of failure to comply with the requirements, as classified in Table II, when tested as specified in 4.4.3.

TABLE II

Defect	Classification	Code No.
Incorrect color (see dwg. 9233945)	Critical	08007
Delay time functioning interval less than minimum (see dwg 9233945)	Critical	08008
Signal explodes prior to reaching minimum delay time (see 3.3.1)	Critical	08009
Signal fails to meet minimum burning time requirement (see dwg. 9233957)	Major	08010
Signal explodes after minimum delay time is reached (see 3.3.1)	Major	08011
Signal fails to ignite (see 3.3)	Major	08012

NOTE: The check for visibility (see 3.3.2) is not required on production samples.

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4.3.3.3.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, two hundred (200) assemblies shall be selected for test. The lot shall be rejected if, during the test, a critical defect occurs, or seven (7) or more assemblies exhibit any of the major defects listed in Table II.

4.3.3.3.2 After three (3) consecutive lots have complied with the criteria of 4.3.3.3.1, eighty (80) assemblies shall be selected for test. The lot shall be rejected, if during the test a critical defect occurs, or four (4) or more assemblies exhibit any of the major defects listed in Table II.

4.3.3.4 Leakage of bag (see 3.4), Major defect, Code No. 09008.-A sample of thirteen (13) packed and sealed barrier bags shall be selected from each lot for this test. If two (2) or more bags show evidence of leakage during test or when bag is opened, the lot shall be rejected. The test shall be performed as specified in 4.4.4.

4.3.3.5 Heat seal of packing bags (see dwg. 9242362), Major defect, Code No. 09009.-Thirteen (13) sealed packing bags shall be selected from each lot for test. If any heat seal fails to comply with the requirements specified, the lot shall be rejected. The test shall be performed as specified in 4.4.5.

4.3.4 Inspection equipment.-Equipment Tabulation Number ET 9233945 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.3.5 Dimensional control of molded parts.-In addition to the normal sampling associated with the Classification of Defects, a sample of at least three (3) parts (as molded) from each cavity shall be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity beginning with the first pieces produced in regular production and after production of each 2500 parts or at the end of each week, whichever occurs first. Individual cavity identification must be provided for each mold. If any defective parts are found during qualification of the cavity, the cavity producing the defective parts will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 Percent for each minor defect. All molds returned from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4. The following subparagraphs identify the items and the minimum inspections subject to the requirements of this paragraph.

4.3.5.1 Cover, housing (see dwg. 9233983)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Large diameter	Gage	12001
102.	Width of cover extension for igniter assembly cavity	Gage	12002
103.	Thickness	Gage	12003

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Minor:			
201.	Height of raised letter, min.	Gage	12004

4.3.5.2 Housing, flare (see dwg. 9233949)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Depth of igniter assembly cavity	Gage	13001
102.	Diameter of igniter cup cavity	Gage	13002
103.	Depth of igniter cup cavity, min	Gage	13003
104.	Diameter of pull wire hole	Gage	13004
105.	Depth of illuminant assembly cavity	Gage	13005
106.	Diameter of flange	Gage	13006
107.	Outside diameter below flange	Gage	13007
108.	Inside diameter of illuminant assembly cavity	Gage	13008
109.	Diameter of pull ring seat	Gage	13009
110.	Chamfer missing	Visual	13010
Minor:			
201.	Total length	Gage	13011
202.	Length to pull ring seat	Gage	1 3012
203.	Width of ignitacord slot, min	Gage	13013
204.	Depth of ignitacord slot, min	Gage	13014
205.	Pull ring location groove missing	Visual	13015

4.3.5.3 Ring, pull (see dwg. 9233948)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Outside diameter	Gage	14001

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102.	Inside diameter	Gage	14002

Minor:

201.	Thickness	Gage	14003
202.	Location tab missing	Visual	14004

4.4 Test methods and procedures

4.4.1 Moisture content

4.4.1.1 First fire composition.-The moisture content of the first fire composition shall be determined in accordance with MIL-STD-1234, Method 101.3. The special solvent to be used shall be equal parts by volume of dimethylformamide methyl alcohol and anhydrous methanol.

4.4.1.2 Illuminant composition.-The moisture content of the illuminant composition shall be determined in accordance with the Karl Fischer Method as stated in MIL-STD-1234, Method 101.2 up to paragraph 5.3. A 50 ml. sample of 1:1 dimethylformamide methyl alcohol solution shall be added into a 250 ml. Erlenmeyer flask (fitted with a rubber stopper) containing 10 grams of the pyrotechnic sample. (Exercise care to exclude atmospheric moisture). The flask shall be stoppered and shaken vigorously to thoroughly mix solution and pyrotechnic sample. Allow the solution to sit 5 to 15 minutes. A blank sample of the 1:1 dimethylformamide methanol solution shall be prepared as described above. A 60 ml. portion of methanol shall be transferred through the sample porthole into the dry titration beaker containing a plastic coated magnet. Stopper at once. The magnetic stirrer shall be started and the water back-titrated in this solvent by adding a slight excess of Karl Fischer reagent to produce a reddish-brown color which persists for a least half a minute. The solution shall be titrated slowly with the water-in-methanol solution until the exact preliminary end point is reached. All succeeding operations shall be completed within 3.5 - 0.25 minutes. 25 ml. of the supernatant solution shall be rapidly introduced, accurately pipetted from the supernantant liquid into the beaker through the same porthole of the rubber stopper. The porthole shall be closed immediately and the contents of the beaker stirred for thirty seconds by means of the magnetic stirrer. A slight excess (10-15 mls) of the Karl Fischer reagent shall be

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added and carefully back-titrated with the standard water-in-methanol solution to the final and exact end point. A water determination of the dimethylformamide methanol blank solution shall be conducted as specified above for the extraction of water from the sample. The percent water in the pyrotechnic sample shall be calculated as follows:

Calculation: $$\frac{2(A_s - A_b)f}{W\ (1000)} \times 100 = \%\ \text{moisture in sample}$$

Where:

A_s = Volume K.F. reagent to titrate sample
A_b = Volume KF reagent to titrate blank
f = Standardization factor (mg/ml)
W = Sample weight

4.4.1.3 Alternate method for first fire composition moisture and illuminant composition moisture.-The moisture content of the first fire composition and illuminant composition shall be determined in accordance with Method T101.4, MIL-STD-286. The temperature setting shall be 100 degrees Centigrade and the time setting 30 minutes. The sample size shall be between 0.40 and 0.60 grams.

4.4.1.4 Paper and paper products.-The moisture content shall be determined in accordance with Method 102.1 of MIL-STD-1234, except that a ten (10) gram sample of each paper product shall be used. The sample shall be dried for two (2) hours.

4.4.2 Candlepower.-The illuminant assemblies shall be supported in a suitable fixture with the ignition end up and ignited with a ignitacord. The candlepower value shall be determined and recorded in accordance with the instructions in 4.3.4.

4.4.3 Functioning.-This test shall be performed at an approved Government facility. Each assembly, after removal of the safety clip, shall be assembled to a suitable test fixture as specified in 4.3.4. The signal shall be functioned by grasping the pull ring and pulling down sharply to remove friction wire. Observation shall be made for compliance with the requirements.

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4.4.3.1 Proving ground tests.-These tests shall be conducted at a Government Proving Ground in accordance with the Aberdeen Proving Ground Acceptance Test Procedure, SI-GR-15.

4.4.3.2 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

4.4.4 Quick leak test (Vacuum Differential Method).-The packed and sealed bags, after conditioning at ambient temperature (70 degrees plus or minus 5 degrees F.) for at least four (4) hours, shall be tested for leakage by submerging in water contained in a vacuum vessel. A vacuum differential of 216 mm (8 1/2 inches) of mercury from ambient conditions shall be drawn and held for thirty (30) seconds minimum (min.) while observing for leakage of air. A steady stream or recurring succession of bubbles from any surface or seam shall be cause for rejection. Where there is no emission of bubbles observed, the package shall be dried, opened, and inspected. Evidence of water within the bag shall also be cause for rejection.

4.4.5 Heat seal of packing bag.-This test shall be performed in accordance with the procedure specified in MIL-P-116. After test the seal shall be examined for compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.-The signal assemblies shall be packaged in accordance with dwg. 9242362.

5.2 Packing

5.2.1 Level A.-The packaged signal assemblies shall be packed in accordance with dwg. 9242363.

5.3 Marking.-The marking of the packaged and packed signal assemblies shall be as specified on dwgs. 9242362 and 9242363.

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6. NOTES

6.1 Ordering date.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock Numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

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6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of

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defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-M. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract, submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Inspection lot size.-It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Proving Ground Test Summary:

	Sample Size	Test	Requirements
First Article	50 of each color	Functioning	See Table II
Regular Production (First three consecutive acceptable lots)	200	Functioning	See Table II
Remaining	80	Functioning	See Table II

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1370-A-285

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